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Final Project: Predicting the 2026 FIFA World Cup Winner

Introduction

With the 2026 FIFA World Cup set to begin on June 11th, this study aims to determine which national team is most likely to secure the championship. As a major soccer fan and someone who will be tuning in to watch the upcoming tournament, I am excited to see if the data supports a potential title win for my favorite team, England. Thus, by analyzing historical tournament data and previous team performances, the project will specifically investigate the statistical probability of England winning the title.

Methods

For this project I will be gathering my data from the online website, Kaggle. This dataset called Soccer World Cup 2022 Prediction, was created by Shi Long Zhuang. For the purpose of this project, I will only be working with the world_cups.csv dataset file. This data file contains every winner, runners-up, third and fourth place finishers of each World Cup from 1930 to the most recent tournament in 2022. In addition, the dataset also includes the host country, goals scored, qualified teams, and matches played by each team. I would also like to note that for this dataset, I will not drop any of the existing columns because they are relevant for the completion of this project.

Model and Approach

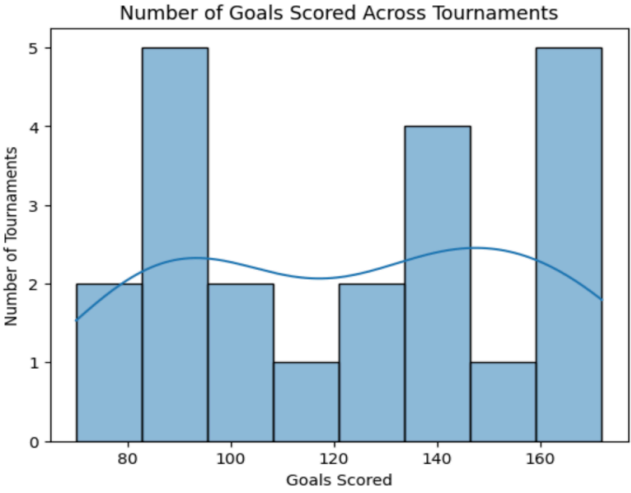
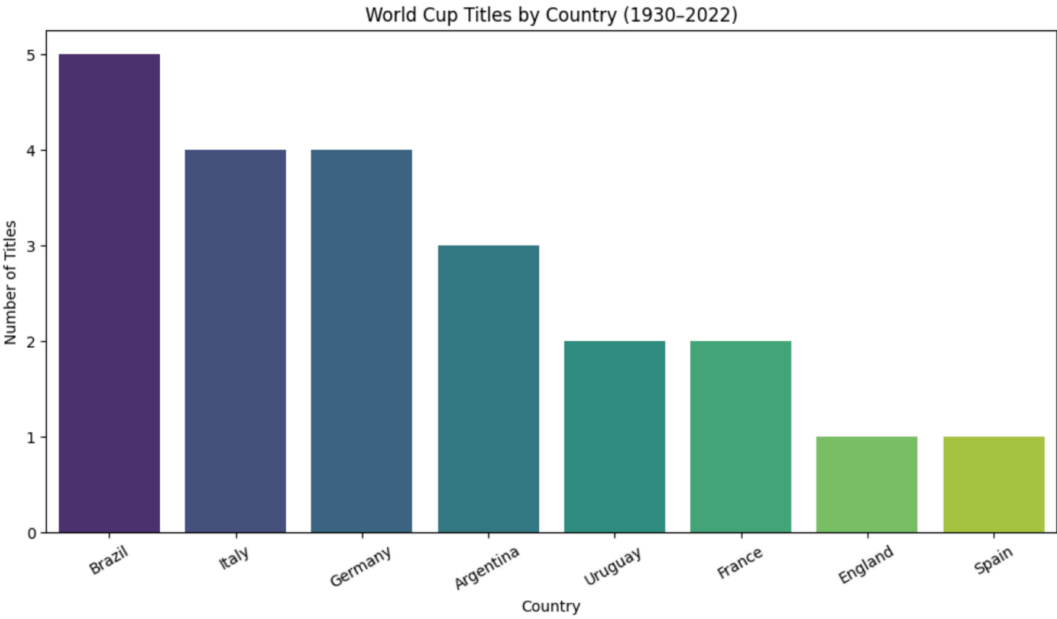
The primary question that I will be focusing on during the entire duration of this project is: “based on the data, which nation is most likely to win the 2026 World Cup?” Other questions that I would also be considering are: “what is the likelihood of the English national team winning?” and “who are the predicted runner-up, third-place, and fourth-place finishers?” Another question that I am considering is how should the model deal with outliers such as countries that no longer exist.

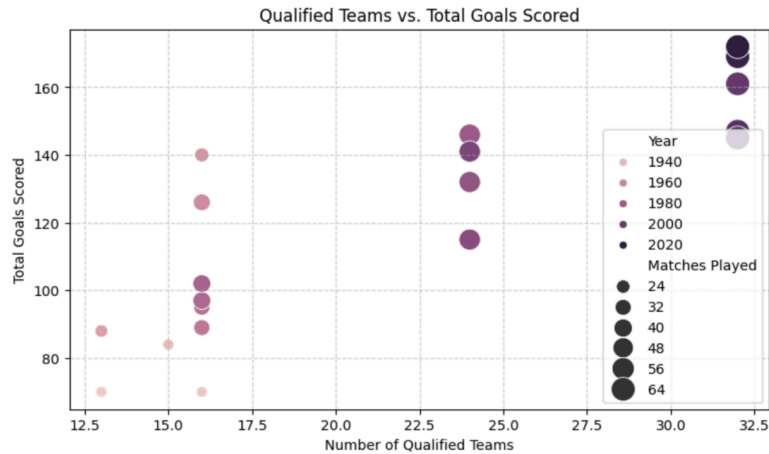
I intend on treating the dependent variable (DV) as the winner column. I plan on handling the data by reshaping it into a binary format where 1 represents a win and 0 represents a loss. For the independent variable (IV), I will be considering the following: host country, runners-up, third and fourth place finishers, year, and qualified teams. For the host country, its role is to calculate the home team advantage by creating a boolean variable. For the previous performers, it will calculate each time a country won, made the runner-up, third or fourth place finish. For the year, its role is to account for recency bias. It should prioritize current team standings over results that are outdated. Lastly, for the qualified teams, its role is to calculate competition density across different tournament eras as the number of participants have increased. On the other hand, I intend on utilizing classification models, Logistic Regression and Random Forest to determine the probability of a specific country winning the tournament. To evaluate the effectiveness of these models, I will utilize the Confusion Matrix to distinguish between true and false predictions. I will also calculate the overall Accuracy Score to test the model’s precision. Additionally, the results will be displayed and evaluated through bar charts and histograms. Note, I will be using AI to assist with the prediction models and data cleaning process.

Results

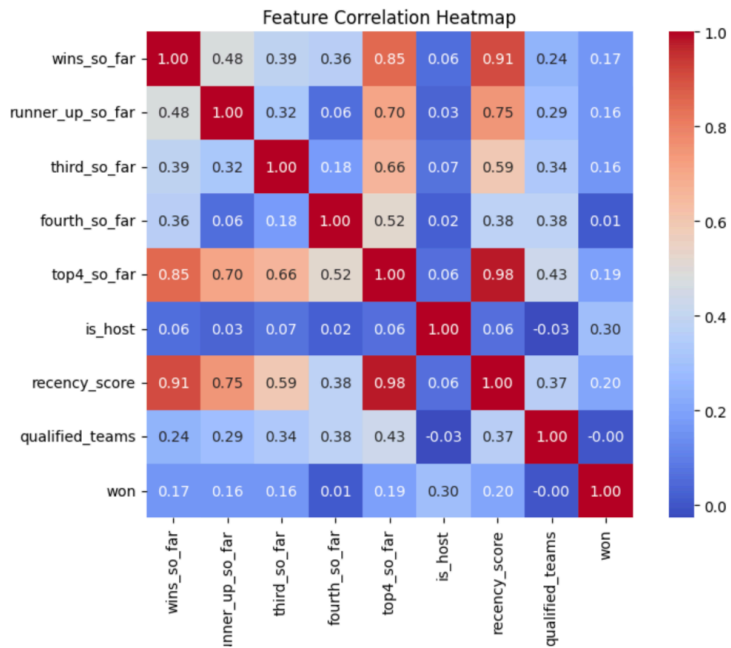
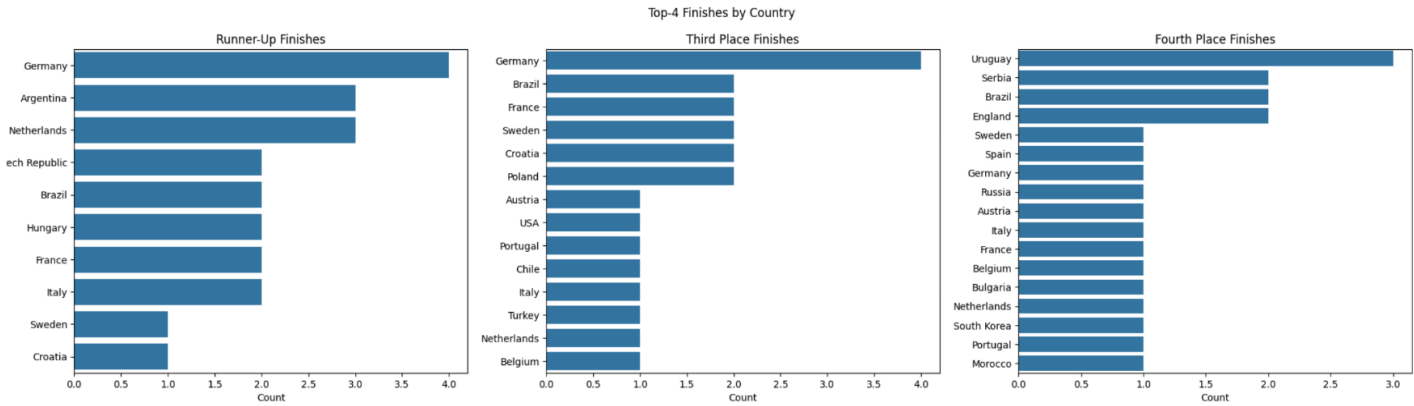
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22 entries, 0 to 21
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  -
0    Year        22 non-null    int64
1    Host Country 22 non-null    object
2    Winner      22 non-null    object
3    Runners-Up  22 non-null    object
4    Third       22 non-null    object
5    Fourth      22 non-null    object
6    Goals Scored 22 non-null    int64
7    Qualified Teams 22 non-null    int64
8    Matches Played 22 non-null    int64
dtypes: int64(4), object(5)
memory usage: 1.7+ KB
```

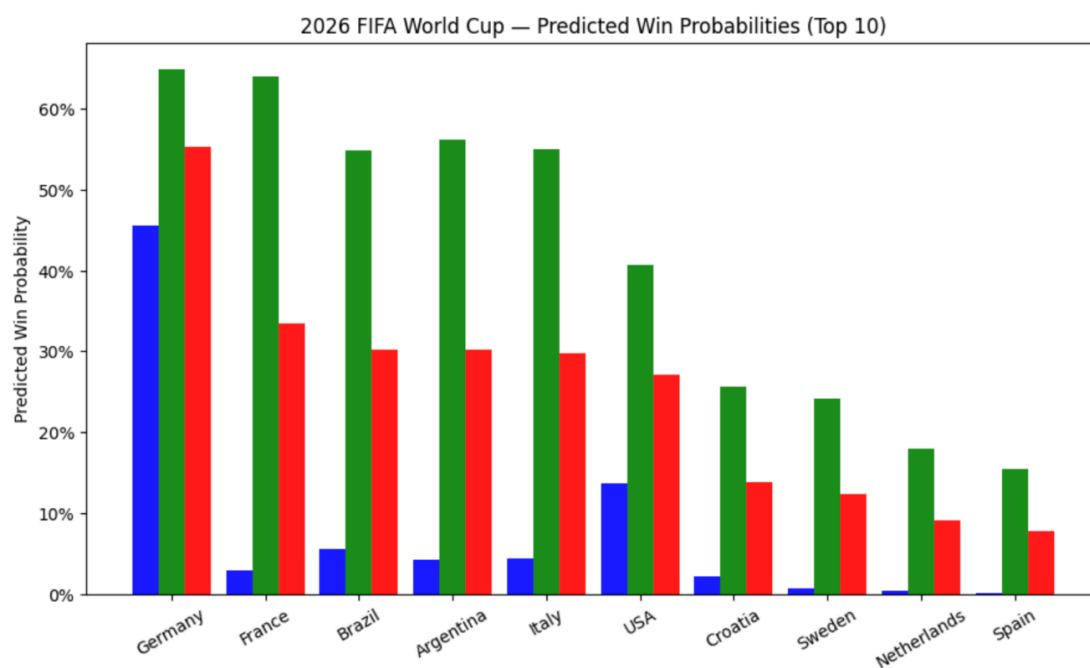
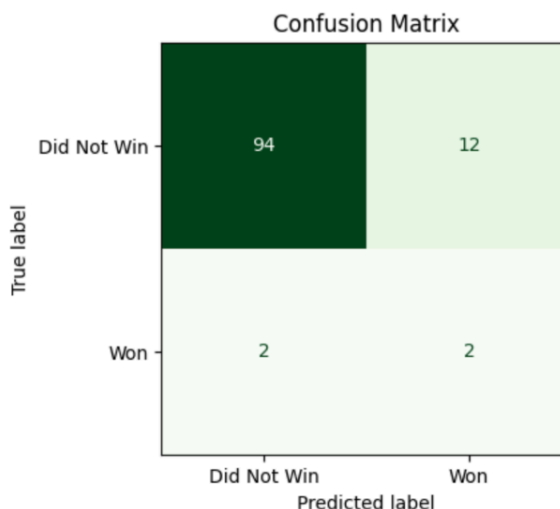
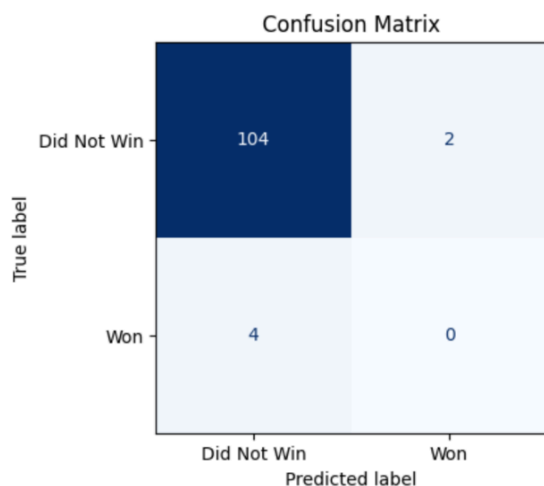
	Year	Goals Scored	Qualified Teams	Matches Played
count	22.000000	22.000000	22.000000	22.000000
mean	1978.909091	123.636364	22.227273	43.818182
std	27.738419	34.841882	7.602830	17.632860
min	1930.000000	70.000000	13.000000	17.000000
25%	1959.000000	90.500000	16.000000	32.000000
50%	1980.000000	129.000000	20.000000	45.000000
75%	2001.000000	146.750000	32.000000	64.000000
max	2022.000000	172.000000	32.000000	64.000000





Pearson correlation between Qualified Teams and Goals Scored: 0.890





2026 FIFA World Cup — Predicted Final Standings

🏆 Winner: Germany (55.21%)

🥈 Runner-Up: France (33.42%)

🥉 Third Place: Brazil (30.25%)

Fourth Place: Argentina (30.16%)

Based on the results, both models were trained on a modified dataset of World Cup tournaments from (1930 - 2022). The feature variables represented cumulative wins, runner-up, top-4 finishes, host-country advantage, recency-weighted performance, and competition density. Overall, it predicted Germany to win the tournament with France as the runner-up, Brazil in third place, and Argentina taking fourth place. Additionally, England's prediction of winning this tournament was 11th, which is the lower tier of title contenders.

The analysis confirms that the selected independent variables (IVs), specifically `recency_score`, `is_host`, and `wins_so_far` were strong predictors of the dependent variable (DV). Both models confirmed that past champions are the strongest indicators for future winners. The correlation heatmap indicated that historical dominance is the most significant factor, but the inclusion of a recency bias was essential in accounting for the decline of historically strong teams and the rise of new contenders. The Home Team Advantage boolean also contributed positively to my findings and it was consistent with the EDA finding that host countries have won ~27% of the time. Moreover, `qualified_teams` had a slightly negative effect. This reflected that the more teams added to the tournament, the harder it is to win.

Both models achieved high accuracy, which is expected given the dataset is heavily imbalanced. For Logistic Regression, it provided a clear, probabilistic baseline. Its strength lies in its ability to be easily interpreted. This allows us to see exactly how much a single win increases a team's future odds. However, its accuracy was slightly lower than the Random Forest, as it struggled with tournament upsets. On the other hand, Random Forest was the more superior model for this dataset. It successfully represented complex interactions between variables, such as the correlation between being a host nation and high density of qualified teams. The

Confusion Matrix for Random Forest also showed a higher rate of True Positives. This means it was better at identifying the actual winner amidst a sea of non-winners.

Thus, the models predicted Brazil and Germany as the top title contenders, followed by France and Argentina. This aligns with historical data and strong performances of the nations in more recent tournaments. Unfortunately, England was predicted in the lower tier of title contenders. This reflects their single World Cup 1966 title and a mixture of decent performances in recent tournaments that have significantly improved their recency score.

Conclusion

The dataset provided a solid foundation, but its primary limitation was its size (22 rows). With only 22 tournaments, the data limited the statistical power of the models. To address this, I had to manipulate the data in order to expand the dataset into per-country rows, which increased the functionality. Additionally, I had to handle outliers such as Germany FR and the Soviet Union. This required manually remapping these countries to their modern equivalents so that the data was concise and accurate. The dataset also does not include team-level stats (FIFA rankings, current squad strength, and qualifying records), which would significantly improve predictions. Moreover, the models predict Brazil and Germany as the top title contenders for the 2026 World Cup, followed closely by Argentina and France. Regarding the research question about England, the data suggests they are in the lower tier of contenders. While their only title was in 1966, their high recency_score significantly boosts their probability. For future work, there are many suggestions that I can make that will further improve this project. For example, I can incorporate current FIFA rankings for each team. This will provide real time strength metrics that historical data can not provide. I can also add player statistics such as average squad age, number of

Champions League and club team winning players. I can add qualifying performance which would provide a more accurate reflection of the team's form. Thus, even though my model did not predict England to win the World Cup, I am still looking forward to seeing if England can come home with a win.

Work Cited

Zhuang, Shi Long. "Soccer World Cup 2022 Prediction." *Kaggle*, 2023,
<https://www.kaggle.com/datasets/shilongzhuang/soccer-world-cup-challenge>. Accessed
16 April 2026.